

CULTURALLY SITUATED EMBODIED DESIGN: A COGNITIVE–ANTHROPOLOGICAL INVESTIGATION OF BALINESE SPATIAL PRACTICE TOWARD GROUNDING GEOMETRY

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Our design-based research project, serving Indonesian students, seeks to ground the geometry technique of auxiliary construction in Balinese cultural funds of knowledge. The ideation phase used an emerging methodological framework for conducting culturally situated cognitive domain analysis. Here we exemplify the framework by reporting findings from examining topographical, architectural, spiritual, and linguistic facets of local historical practice, and from personal training in traditional dance. Key results include identifying and characterizing a diagrammatic structure—a concentric eight-rayed matrix—as a thematic spatial-orientation form tacitly underlying heritage practices. We explain and demonstrate how our proposed design, Geometry Resources in Dance (GRiD), taps and surfaces this covert structure as an epistemic resource offering young students culturally authentic enactive grounds for mathematical learning.

Keywords: Cognition, Culturally Relevant Pedagogy, Design Experiments, Geometry and Spatial Reasoning

Constructivist frameworks aspire to guide the design of educational activities in which students draw on their prior knowledge to learn targeted subject matter content (von Glasersfeld, 1987). Yet this prior knowledge that students bring to the activity may vary across cultural contexts. *How do we create a constructivist learning activity for a child living on our antipodes?*

This paper emanates from a US-located design-based research project evaluating a novel activity for Indonesian students to develop the geometry skill of constructing auxiliary lines. Previous publications from this ongoing project are empirical papers reporting the effects of the design on students' spontaneous habituation into the practice of imagining and marking lines as their semiotic means of supporting their collaborative geometrical argumentation (Apsari & Abrahamson, 2024). The current paper, however, is essentially methodological, and so we will step back in time from the empirical evaluation stage of our project to share our emerging insights concerning its early ideation stage. The core of our insight concerns the crucial function of members of the target community as lead- or co-designers of culturally situated learning activities.

In and of itself, advocating for co-design is hardly new (e.g., Vandekerckhove et al., 2023), the common theme being that prospective users bear authentic expertise that the designers may otherwise hardly consider, if at all. The epistemic quality of this authentic local expertise, however, has been insufficiently characterized, we believe, and consequently its contributions to the co-design process may be inadvertently delimited. Namely, authentic expertise may not always be explicitly articulated but, rather, be implicit to mundane practice. As such, our general methodological problem concerns the question of tapping the authentic expertise of local co-design stakeholders. We submit that *in creating culturally situated mathematics learning activities, members of the community may bring to bear tacit knowledge which an exogenous researcher might never fathom*. The rationale of this paper is to narrate the early phase of our design process, during which Author Apsari—a member of the target community envisioned the activity's core artifact. Only later in retrospect, through an inquiry process we call *cognitive–*

anthropological domain analysis (cf. Eglash, 2005; Verran, 2001), did we realize that the artifact she built is culturally situated in multifarious aspects of local heritage practice. Our findings have led us to believe that Apsari's initial ideation was not happenstance. Rather, it drew on deep wells of tacit cultural–historical knowledge from whence she conjectured a form she didn't know she knew.

The paper opens with an overview of our design rationale for situating mathematics in movement practice. Then, we present findings from our ideation stage, in which we applied cognitive–anthropological and participant-ethnographic methods to probe for cultural–historical roots of Balinese movement practices, particularly dance, that could potentially ground the geometry skill of building auxiliary constructions. Next, we describe the artifacts and embodied activities we designed to anchor students' mathematical experiences to their heritage practices. Finally, we provide a case study illustrating how these activities elicit and leverage young Balinese dancers' proto-mathematical tacit perceptual strategies. Through this work, we contribute to the field's growing discourse evaluating the purchase of embodiment philosophy on the theory and practice of mathematics education, while advocating for pedagogies that foster and affirm students' cultural heritage. We are thus exploring culturally situated embodied design.

Design Rationale: Grounding Geometry in Movement Practice

Auxiliary lines are vital cognitive tools for solving geometry problems, because they invite alternative perceptual constructions of the shapes in question (Palatnik & Dreyfus, 2019; Polya, 1957). However, teaching and learning how to construct auxiliary lines remains challenging, because students are expected to “see” and then build something that, by definition, is not yet in their visual field; moreover, pointing out to students where to draw these lines could defeat the purpose of training them to do so independently (Fan et al., 2017; Gridos et al., 2022). Addressing this problem of practice, our design-based project that drawing on the action-based embodied design (Abrahamson, 2014) seeks to identify or create a culturally situated context wherein students would intuitively construct auxiliary lines.

By leveraging students' heritage dance practices as their cognitive–affective grounds for understanding curricular subject matter, we draw on students' *funds of knowledge*—the everyday and community-based situated know-how they acquire through participating in the mundane enactment of cultural practice (Moll et al., 1992). In doing so, we strive to avoid the ethical and pedagogical pitfalls of unilaterally imposing on students hegemonic ontologies that conflict with their primary intuitions (see also Bamberger & diSessa, 2003). Instead, we aim to craft lessons that inspire students to embrace and affirm their cultural identities (Ladson-Billings, 1995).

Our initial inquiry focused on dance, because dancers are known to create imaginary figurative elements to navigate space and enhance the execution of movements (Clark & Ando, 2014; Movement Research, 2012). From a resonant enactivist perspective, Hutto & Sánchez-García (2015) introduced the construct of *attentional anchors*—perceptual resources, such as imaginary linear constructions, that individuals project into their sensory field to coordinate the performance of challenging movements. For example, a dancer in training used a triangular pattern she discerned on a floor mat and then “carried” it with her as she moved, to perform a choreography sequence (Apsari & Abrahamson, in press). We reasoned that this intuitive perceptual heuristic of conjuring imaginary lines to mobilize task performance, which has been documented in dance practice, is precisely what we are looking for in geometry practice. As such, a propitious activity would solicit from students attentional anchors that, given an appropriate dance-cum-geometry context, would materialize as auxiliary lines. The question then became how to situate geometry tasks in the dance context.

Our inquiry is rooted in Balinese dance, a tradition Apsari has practiced since childhood. Her emic identity intersection as both a Balinese dancer and mathematics educator uniquely positions her as a design-based researcher who advocates for embodied mathematics activities rooted in Balinese dancing traditions. Therefore, our design was created *by* and *for* the Balinese community to experience mathematics in the context of their own cultural heritage.

The Hidden Lines of Balinese Dance: Unearthing the Geo-Cosmological Grounds of a Cultural–Historical Spatial Discourse

This section overviews the cultural settings of our design-based research project by introducing the cultural–pedagogical context of Balinese traditional dance practice, highlighting the cognitive challenges of coordinating its movement sequences. We explain dancers’ common solutions for enacting these movements by way of tacitly projecting imaginary lines onto their perceptual horizons. We then focus on an apparent paradox: whereas these perceptual tactics are prevalent among Balinese dancers and, moreover, the imaginary lines configure an apparently identical form across the dancers, the form is not deliberately or explicitly instructed. We explore this paradox by asking whether the form draws on deeper cultural roots, and our exploration will reveal a spatial form—an eight-rayed matrix—apparent across topological, cardinal, architectural, spiritual, and linguistic aspects in Balinese collective psyche and spatial practice.

In Balinese culture, dance is not merely a form of artistic expression but a practice that is deeply embedded within social and ceremonial life (Dibia & Ballinger, 2023). Young members of the community are inducted into dance practice through their involvement in the tradition, beginning with observation and gradually transitioning to active engagement in activities of increasing complexity (Paradise & Rogoff, 2009). In such environments, learning is a lifelong engaged process, where individuals are enculturated into their participatory roles as active community members (Lave & Wenger, 1991).

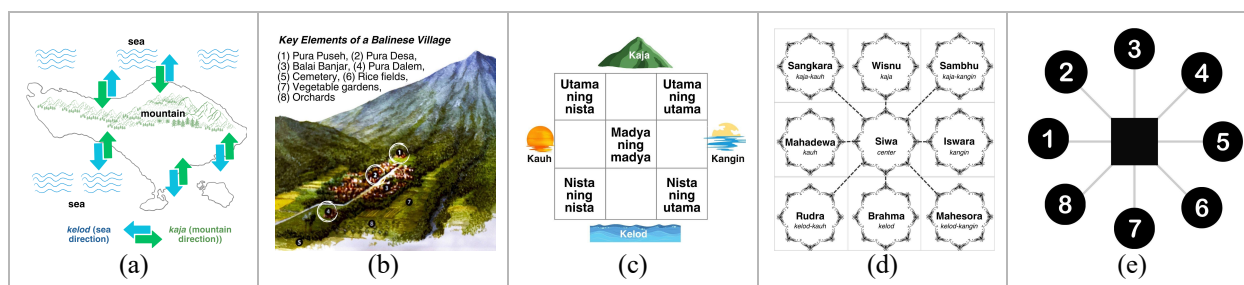
Balinese dance is traditionally taught in community settings, where it is mediated intergenerationally. Unlike formal schooling, Balinese enculturation into traditional dance practice does not follow any set curriculum, rather, it is contextual, ad hoc, and responsive (Ingold, 2000; Lindwall & Ekström, 2012; Paradise & Rogoff, 2009), oriented on imbuing bodily actions with narrative meanings. Similar to the case of European ballet, Balinese dance instruction begins with basic postures constituting the genre’s choreographic vocabulary.

The foundational posture of Balinese dance, *agem*, is an asymmetrical body organization that represents imbalance (Dibia & Ballinger, 2023). In “right *agem*,” the left foot is positioned the distance of one fist in front of the right foot, the body weight is on the right foot, the shoulder blades are squeezed together, and the arms bent with the wrist lower than the elbow. Also, the right hand is placed at eye level while the left hand is placed at chest level, and the head is tilted slightly to the right. For the left *agem*, the entire right–left configuration is switched. To perform a dance, one should master the *agem* and hence must find balance within this imbalance posture.

Finding balance is challenging for a beginner dancer. In response, the teacher usually instructs students to establish an imaginary vertical line, running down from the sky/heaven, through their body, and down to the underworld. This tripartite corporeal axial extension—heaven, body, underworld—implements the Balinese concept of *Tri Angga*, by which everything can be divided into three parts: *utama* (high), *madya* (middle), and *nista* (low). The *utama* of the universe is the zenith, such as the heaven, sky, or mountain, the sacred dwelling place of the Gods as well as the spirits of ancestors. Humans live on the earth, the *madya* section. The *nista* comprises lower geo-cosmological echelons, such as the sea or underworld, where lie malevolent spirits and the essence of impurity (Budihardjo, 1995; Davidson, 2023).

Reflexively, the human body itself constitutes the tripartite moralistic–cosmological diagram, whereby the head is *utama*, the torso is *madya*, and the feet are *nista*. Mythological in its cultural roots, *Tri Angga* runs deep in contemporary Balinese discourse and practice, implicitly suffusing variform aspects of praxis, ethos, and mores, including the topographic layout of villages and, fractally, within the village, the architectural design of each traditional home.

Figure 1: Balinese Geospatial Matrix Correlated with Divine Mythic Cosmology



Note. (a) The mountains area of Bali are mostly in the shaded region, with the highest and the most sacred mountain being Mount Agung in East Bali (adapted from Budihardjo, 1995, p.44). The modern, loose translation of *kaja* is “north” and *kelod* is “south.” These meanings are reversed for dwellers of the Buleleng region in North Bali. (b) Topological layout of a village on the foothills of Mount Agung. The circled numbers 1, 2, and 4 mark the locations of the village’s three main temples (adapted from (Davidson, 2023, p.7). The temple of origin (Pura Puseh) and the temple of creation (Pura Desa) are located in (1) and (2), respectively, towards the mountain (*kaja*), while the temple of death (Pura Dalem) is located in the village’s *kelod* sector (4) near the cemetery, which is considered impure. (c) The *Sanga Mandala* nine-celled spatial grid is implemented as the Balinese prevailing architectural principle that valorizes *kaja* over *kelod* and *kangin* over *kauh*. The *kaja–kelod* and *kangin–kauh* axes respectively imbue the space with nine totem/taboo gradients, thus implicating the *utama ning utama* (sacrosanctum) at the *kaja–kangin* (towards the mountain and the sun rise) grid. (d) Every position in the *Sanga Mandala* corresponds to the topographical–cardinal value-laden cartographic matrix that intrinsically bespeaks the *Dewata Nawa Sanga* deific system, here depicted in corresponding diagrammatic form. (e) An illustration of mandatory orientation points was created by a renowned Balinese dancer and teacher to illustrate how he centered himself and navigated the space while dancing; the numbers capture his enumeration during the reflection conversation, with Apsari, following dance class.

The plurality, if not majority of Balinese villages are configured on the topographical cardinalities of *kaja* (towards the mountain) and *kelod* (towards the sea) axis (see Figures 1a&b). In addition to the *kaja–kelod* axis, Balinese also use the *kangin–kauh* axis corresponding to the sun’s orientations. *Kangin* is where the sun rises, roughly east, for near-equatorial Bali, and *kauh* is where the sun sets, correspondingly the west. As in the case of *kaja–kelod*, the wide-ranging *kangin–kauh* axis, too, serves in partitioning a local territory into three regions. Together, the tripartite *kaja–kelod* division and tripartite *kangin–kauh* division factor a nine-celled cross-product spatial matrix known as the *Sanga Mandala* (see Figure 1c), a sacred compass consisting of a center from which emanate eight orientations, per the *Dewata Nawa Sanga* (Figure 1d) principle (*Dewata*: gods, deities; *Nawa* and *Sanga* both mean nine). Balinese people associate each compass direction with a particular deity along with all it portends. Traditional Balinese architecture, whether a sacred temple structure or a humble dwelling compound, is organized based on this *Sanga Mandala* spatial–deific concept. The holiest direction (*utama ning utama*) will be both in the direction of the mountain (*kaja*) and towards the sun rise (*kangin*), hence *kaja–kangin*. In the typical Balinese house, the *kaja–kangin* sector is reserved for the family temple. On the other hand, the impurest direction (*nista ning nista*) is towards the sea/ocean (*kelod*) and the sunset (*kauh*), hence *kelod–kauh*, where the kitchen is commonly placed.

Bandem & deBoer (1995) illustrated how the *kaja–kelod* axis could also be discerned in determining locations of Balinese dance performance. In general, Balinese dance can be divided into three categories. First, dances of the most sacred category (*wali*) are to be performed in the inner courtyard of the temple (*jeroan*) and only for religious ceremonial purposes. The second type of dance is the ceremonial performance of *bebali*, which is also ritualistic but not as sacred as *wali* and can be performed in the second courtyard or the middle of the temple (*jaba tengah*). The third and last dance type is the *balih-balihan*, performed in the temple's outer part (*jaba*).

As an interim summary, we witness that the eight-rayed traditional Balinese orientational system configures the spatial bearings of a sentient corporeal epicenter, the Balinese individual in their sociocultural settings. We wondered whether the form could similarly extend to mathematical scholarship, specifically situating geometry studies in cultural–historical authentic practice of movement performance. To begin with, one might ask what function this static spatial matrix might play in the dynamic performance practice of a Balinese dancer. As we discovered, the Balinese geospatial–deific matrix imbues traditional dance practice, too.

Balinese dancers learn to imaginatively partition the stage according to the nine-celled gridded spatial organization and center their performance on its central cell. Moreover, just as a village layout reinstates at a micro-level the land's geo-topography, so the dancer's body recapitulates the *Sanga Mandala*. Finally, the dancer-centric invisible matrix is portable—dancers can translate and rotate the matrix across the stage to serve their evolving coordination needs. To illustrate the incorporation of the deific matrix onto the dancer's position and orientation, we reproduce, below, a translated excerpt from recordings of Apsari's private lessons with one of Bali's most prominent Gambuh dancers and teachers, I Made Suteja.

I Made Suteja: When we are dancing . . . a dancer is . . . for me, being the Siwa. The position of Siwa is in the center.

Siwa (the God Shiva) is the center of the Balinese *Dewa Sanga* diagram (Figure 1d). By “being the Siwa,” Suteja referred to his attempt to center himself vis-à-vis the system of imaginary horizon points he perceives and abides with as he dances. As Suteja explained:

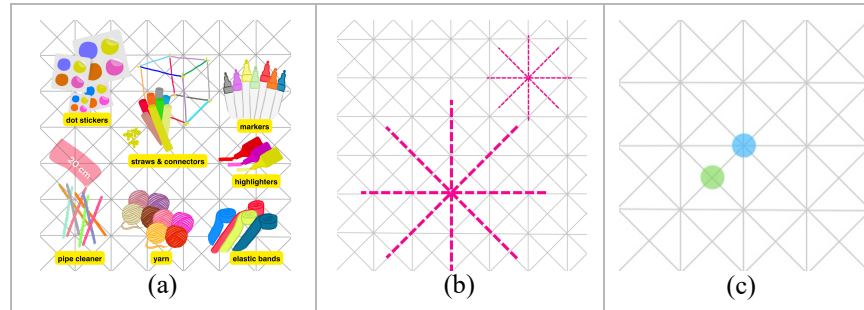
(When) we are dancing, there are eight mandatory points. If we go there [*orients himself toward a location marked with a square in Figure 1e*] . . . the points are . . . the sides, right and left [*points with his right hand to the right and then with the left hand to the left*], straight forward, corners right and left [*points with his right hand to the front, the right corner, and then the left corner*]. That is one, two, three, four, five [*points with his right hand to the directions of 1–5 in Figure 1e*]. On the back . . . six, seven [*points with his right hand to the direction of 6 & 7*] . . . eight [*points with his left hand to the direction of 8*].

While the *Sanga Mandala* and *Dewata Nawa Sanga* forms play explicit roles in Balinese life, their practical function in traditional dance is mostly implicit. Dancing instructors do not explicitly state that students should consider the nine-grid. Notwithstanding, dance teachers are evidently aware of the student's orientation, correcting it when a student moves off course. Likewise, we will bear these forms in mind as we turn to discuss our culturally situated design.

Geometry Resources in Dance (GRiD): Design Overview and Case Study

The objective of the design is to ground embodied activities for geometry learning within culturally authentic dance practices, using low-cost dedicated materials, such that students' spontaneously generated attentional anchors could be objectified as auxiliary lines serving in solving geometry problems. Meeting this objective required devising an appropriate space.

Figure 2: GRiD and Its Schematic Form as Staging the Balinese *Dewata Nawa Sanga*



Note. (a) The components of GRiD, a diagrammatic floor mat with various accessories that can be utilized to facilitate exploration, discovery, and discussion. (b) The crisscross arena of the GRiD layout resonates with the structure of *Dewata Nawa Sanga*, with the eight concentric rays extending indefinitely. (c) Compare the incomplete 4-rayed node and the complete 8-rayed node on the GRiD, marked with green and blue dots, respectively.

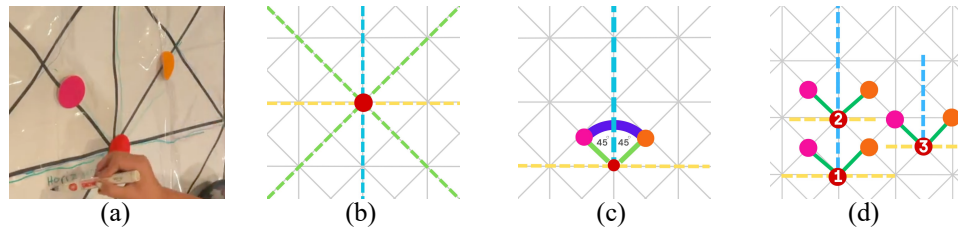
To achieve the objective, we designed a system of resources called *GRiD—Geometry Resource in Dance* (Figure 2a), which comprises a laminated mat and various colorful accessories. GRiD's mat is covered with four sets of parallel lines angled respectively along the vertical, horizontal, and two main diagonal axes. The unit structure of these crisscross lines serves as a schematic reconstruction of the Balinese *Dewata Nawa Sanga* spatial matrix discussed above, with eight compass bearings and one center point (Figure 2b). Students were invited to use GRiD's media to solve our designed activities (see examples in Apsari & Abrahamson, 2024).

GRiD was first piloted with a Grade 5 California-based student who practiced Balinese dance. Once we had analyzed these pilot data and accordingly modified and expanded the activities, we implemented the experimental design with 76 Bali-based Grade 5 students, who identified as having had various levels of experience with Balinese dance. Our microgenetic analysis of these audio–videotaped data focused on detecting what functions GRiD played in students' task performance, problem solving, collaboration, and learning.

As we report in Apsari & Abrahamson (2024) we found that GRiD served the students as a semiotic means of objectifying their presymbolic spatial orientation (cf. Radford, 2003). Namely, GRiD's visible diagrammatic structure enabled students to materialize, articulate, and index their attentional anchors *to form, monitor, and regulate* the Balinese postures and choreographed movement routines. The objective of the current paper is primarily to report on, and evaluate, the project's cognitive–anthropological domain-analysis methodology. Here, we present a single case study to illustrate how GRiD's structure echoes students' tacit perceptual form they may have developed through participating in the Balinese dance tradition. In turn, this case study lends credence to our claim that Apsari *tacitly* recreated the form in GRiD. For empirical support of our claim concerning GRiD's efficacy, readers are referred to Apsari & Abrahamson (2024).

Our activity's first task is to enact on GRiD a basic Balinese-dance feet posture called *Tapak Sirang Pada* (henceforth, TSP), where the heels touch while the feet are rotated out, creating what Euclidian geometry labels as a 90° angle. Our activity protocol does not include any specific instructions as to *where* on the GRiD the students should place their feet. Nevertheless, many of our participants placed their feet such that their heels converged on the concentric eight-rayed intersection (hence, “8-node”; In Figure 2c compare the green circle marking a 4-node and the blue circle marking an 8-node). Anna (pseudonym), a 10-year-old Balinese dancer in training, began the activity by demonstrating TSP.

Figure 3: Anna's Construction of a System of Auxiliary Lines

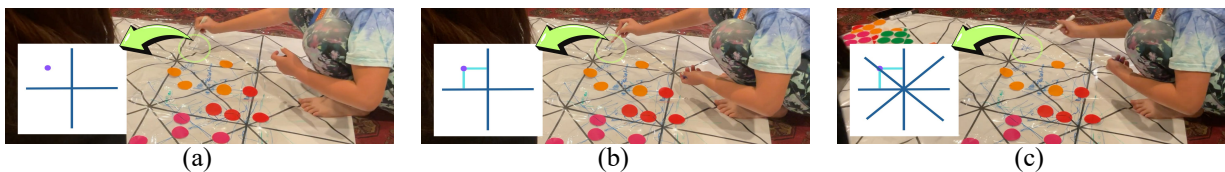


As she stepped on the GRiD mat, Anna immediately positioned her heels on an 8-node. Using a color marker, she then highlighted a horizontal line and a vertical line intersecting at the node, which she then marked with a dot sticker. Later, Anna also marked the tip of her toes with two other dot stickers and connected each of these to the sticker marking her heels' juncture (Figure 4a). Prompted to explain why she marked that spot, Anna replied that the junction point of her heels is "*the place where it's almost like a center where everybody comes to meet.*" By "everybody," she then clarified in words and drawing, she meant all the lines she had marked—horizontal, vertical, and diagonals—to spatially locate her feet per the TSP rule (Figure 3b).

Anna further utilized the lines she had created to reason about the angle measurement of her feet's TSP aperture. First, she divided her stance into two spatial regions, each composing a right angle. Thus, each foot came to constitute a bisector halving each right angle, respectively, resulting in two 45° angles on each side. As such, the angle measurement between her feet, said Anna, equals a 90° angle (Figure 3c). She further argued that when she moved to different locations as per the choreography, e.g., forward, backward, and sideways, she was trying to maintain the preciseness of her TSP by porting the helpful lines with her (Figure 3d). For a more detailed elaboration of Anna's geometric reasoning about the 90° angle, see Apsari & Abrahamson (in press).

According to Anna, the vertical and horizontal lines she employed to evaluate her TSP were the primary lines—she endearingly named them "the main guys" and "the original." In Figure 4 1, where "A" stands for Anna, we will soon observe her schematization of the GRiD's 8-node unit as an intact system that enables her to determine with precision an object's relative position.

Figure 4: Diagonal Lines for a Precise Location



A: But then I feel like diagonals... I wouldn't say they're new, but I feel like they're more like . . . like if this guy was like right here [draws a small-scale Cartesian diagram and marks a point on its second quadrant]. . .

A: They're like it's basically... um like close to the vertical line... [draws a vertical line to connect the point with the horizontal axis and a horizontal line to connect the point with the vertical axis] . . . Close to... um the horizontal line, and that would help.

A: But then I feel like the diagonal lines they're like they help you get more specific. . . . So if I draw my diagonal lines [draws diagonal lines on her Cartesian diagram], you can say like... that dot is on the diagonal line.

Discussion

We have been discussing Balinese dance practice from a cognitive–anthropological perspective that highlighted a specific spatial form apparently cutting across multiple cultural facets and, thus, covertly lending systemic coherence to the phenomenology of being a Balinese individual. Whereas a deeper ethnography would lie outside this paper’s disciplinary scope, we surmise that the Balinese spatial matrix serves its people as a generative cultural codex.

From our cognitive domain analysis, we discovered that learning to dance may be more than physical entrainment in some ad hoc choreographic routines—learning to dance may covertly constitute broader cognitive and epistemic entrainment in Balinese spatial, moral, and spiritual practice. The society’s situated geospatial eight-pronged system of cardinal orientation, which mediates the enactment of mundane cultural practice, such as physical navigation and narrative reference, is embedded in and learned through dance. In dance, the Balinese orientational form serves as a stylized diagrammatization of cultural dynamics. Partaking in dance, the student incorporates the divine cosmology and mythology of the land (cf. Schwabe, 2010, re Rudolf Laban; and see Reed & Bril, 1996, on the enculturation function of the *field of promoted action*). As such, dance instruction mediates subtle intergenerational knowledge.

GRiD instantiates the Balinese divine matrix *Dewata Nawa Sanga*, inviting students to tacitly engage with the diagrammatic structure, which they laminate with their attentional anchors to accomplish their choreographic task, then animate with color highlights to articulate their strategy, whereby the GRiD mat materializes tacit Balinese spatial practice into language.

Our case study illustrated how Anna created auxiliary lines to help her enact and evaluate the TSP in different locations. Our conjecture was further ratified in Figure 4, in which Anna’s reasoning suggested that the GRiD layout affords the Balinese implicit 8-ray schematics centered on the dancer’s body. Anna’s assertion that the horizontal and vertical lines were the primary focus for her, while the diagonal lines enhanced her precision, was particularly intriguing. Krichevets et al. (2014) found that when experts faced problems locating a coordinate point on a Cartesian plane, they often traced the diagonal line where the point lies, instead of triangulating the horizontal–vertical cross-product. The strategy rendered their actions more efficient by reallocating visual resources. Anna, while not yet an expert, developed an idiosyncratic spatial organization that helped her accurately locate an object.

Conclusion

So often, educational design that claims to tap cultural practices from certain groups does this superficially, resulting in ineffective interventions that confuse both teachers and students. Espousing embodiment perspectives enables us to offer a phenomenologically authentic cognitive–anthropological operationalization of the culturally responsive design framework. Culturally situated embodied design activities stand to mediate mathematical skills. Further research as well as conversations in the community of design-based mathematics-education researchers could substantiate our methodological conjecture that native co-designers stand to play critical roles by tacitly bringing to bear their heritage cultural know-how.

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